

Using 2g Servos with a 2S (7.4v) Battery



Moth park flyer running 300mAh 2S battery with 2g servos

Introduction

Many lightweight RC aircraft use tiny 2g servos, but they aren't designed to run directly from a 2S (7.4V) battery. I've tried it and the servos go into a random fit before eventually dying!

In this article, I present a couple of options to get around this. I demonstrate the setup using the popular cost effective FlySky FS-i6 transmitter and AFHDS 2A receivers, although many of the concepts apply to other RC radio systems as well. Get your soldering iron ready and lets go!

Also, check out my video on YouTube on this topic:



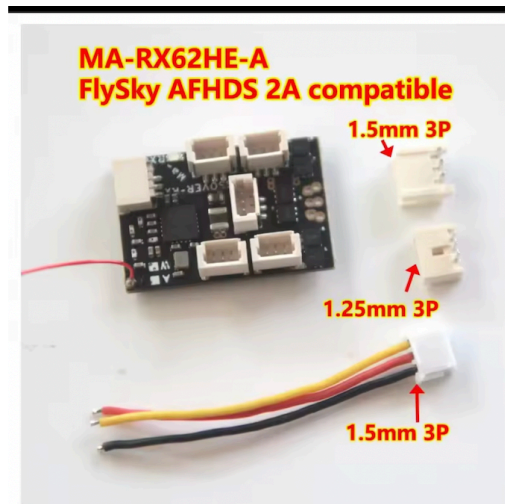
What are the options?

Here are two options for you to consider that will work:

Option 1: Receiver with Built-in ESC and BEC

Purchase a 2S-capable receiver with built in ESC (and BEC), which the servos will plug into directly.

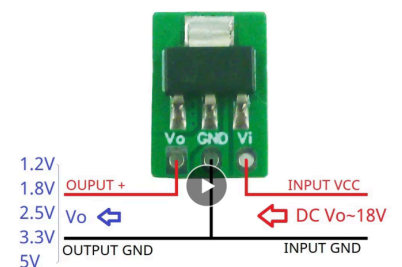
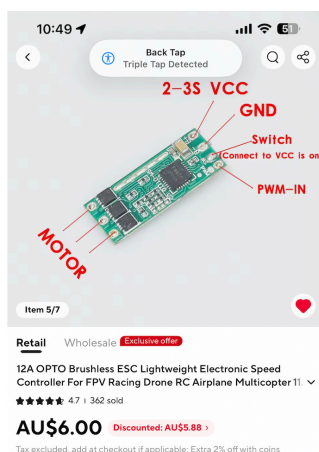
Here is an example of one I have tested:



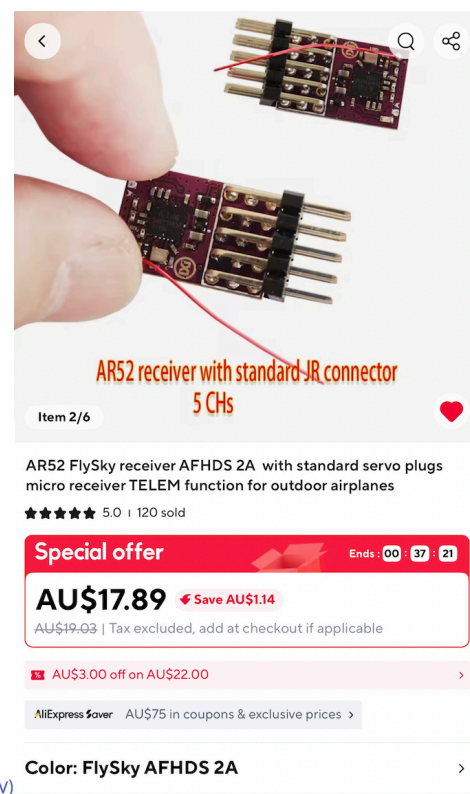
Option 2: Use Voltage Regulator to protect servos

Purchase a basic 2S-capable receiver and separate ESC. Also purchase a 3.3 or 5V voltage regulator. The voltage regulator is used to keep the voltage at 3.3 or 5V for the receiver and servos while the ESC and brushless motor gets 7.4V.

Here is an example of a combination that I have tested:



Dropout Voltage: 80mV (I_{OUT} = 100mA)
Maximum Output Current 800mA (V_{IN} = V_{OUT} + 1.3V)



Technical Details

Option 1: Receiver with Built-in ESC and BEC

Purchase a 2S-capable receiver with built in ESC (and BEC), which the servos will plug into directly.

Pros:

- Guaranteed to work with your setup
- Simpler to implement
- Less soldering

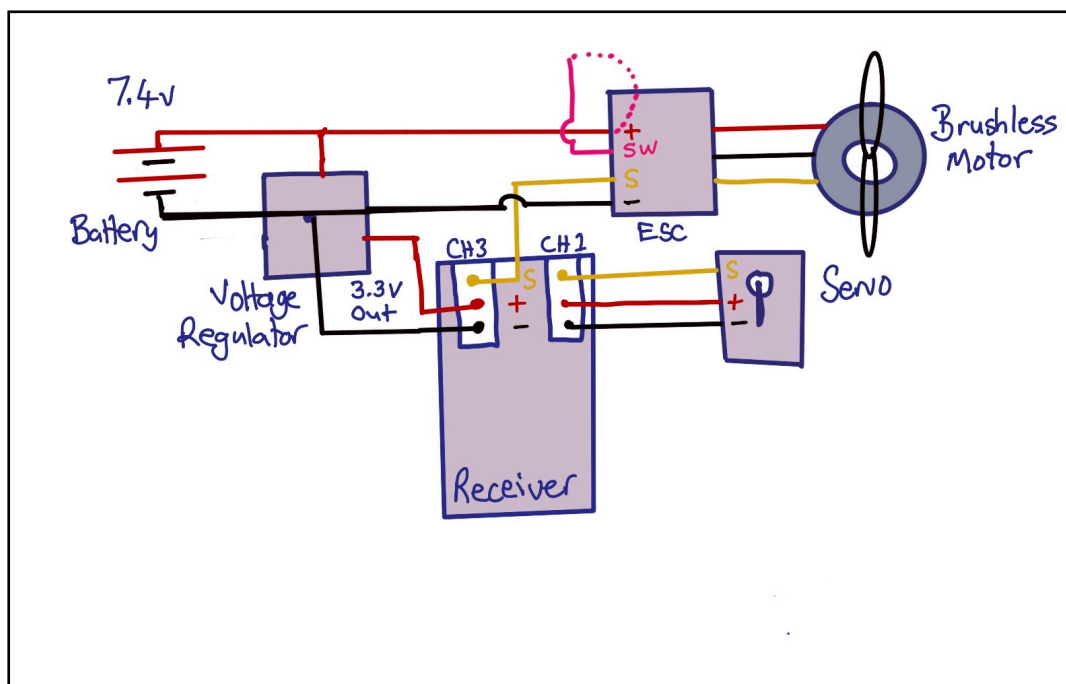
Cons:

- Costlier
- If you blow the onboard ESC, the receiver can only be used in a glider

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This is how to wire it all up:



Pros:

- Much cheaper

Cons:

- Can't guarantee all parts will work together
- A little bit more soldering

Links to Parts discussed in this forum and YouTube video:

Note, these prices (A\$) excluding postage will vary depending on vendor and time of year.

Flysky FS-i6 Transmitter - \$63.69+\$2.36 postage

FLYSKY FS-i6X I6X AFHDS 2A RC Transmitter with X6B IA6B IA10B Receiver 10CH 2.4GHz for Aairplane Helicopter FPV Racing Drones

https://a.aliexpress.com/_mOYtmCF

4-Channel 1-2S Receiver: \$8.09

FS2A 4CH AFHDS 2A Mini Compatible Receiver PWM Output for Flysky i6 i6X i6S / FS-i6 FS-i6X FS-i6S Transmitter

https://a.aliexpress.com/_mNfHWsx

2S ESC: \$4.69

3.3V voltage regulator: 12A OPTO Brushless ESC Lightweight Electronic Speed Controller For FPV Racing Drone RC Airplane Multicopter 1104 1106 1806 Motor

https://a.aliexpress.com/_mPSunLd

Voltage Regulator: \$5.82 for 6x

Tiny Out 1A 1.2V 1.8V 2.5V 3.3V 5V 80MV Drop Voltage 60uA Standby Step-Down Buck LDO Module rep AMS1117 7805 for Arduino ESP32

https://a.aliexpress.com/_mMVEhvt

2g Servos: \$7.03

DSPOWER 2g Micro Digital Coreless Servo Plastic Gear JR/JST Plug Mini Servos for Rc Helicopter Plane Boat Car Model Accessoires

https://a.aliexpress.com/_mOzbonh

Choose one of 3 connectors:

JR2.54 (large), JST1.0 and JST1.25 - needs to match the sockets on your receiver.

For the FS2A receiver above, I used the JR2.54 connectors on my Jump Racer Jnr, which are very robust.

MC1104 Motor: \$9.19

1104 Mini Brushless Motor 14mm 4300kv for 1S-3S RC Drone FPV Racing Quadcopter

https://a.aliexpress.com/_mPIMTof

RS1102 Motor: \$53.69 for 4

4PCS Happymodel RS1102 1102 10000KV 13500KV 2S Brushless Motor 1.5mm Shaft for Mobula7 O4 Micro FPV Racing Drone RC Accessories

https://a.aliexpress.com/_mNSkedz

3016 Propellers: \$12.69 for 2x pairs (for the RS1102 motors)

2 pairs / 6 pairs Gemfan 3016 Propeller 1.5m hole 3 inch 3-Blade CW CCW FPV

Propeller Mini Props For 3inch FPV Racing Drone

https://a.aliexpress.com/_mKrvzWL

5040 Propellers: \$7.72 for 5x (for the MC1104 motors)

5PCS 5030 6035 7035 8040 8060 9050 1060 1160 Propeller For RC Airplane

https://a.aliexpress.com/_mrhoMXt

7A 2S/5A 3S Receiver with onboard ESC, BEC and Servo sockets:

\$38.79 Ma-RX62HE-A FLYSKY AFHDS 2A 6CH Micro Receiver Built-in 7A/2S(5A/3S)

Brushless ESC for FLYSKY Radio Transmitter

Select 1.5, 1.25, 1.5mm 3P connectors

https://a.aliexpress.com/_mrr6tTl

I've used this on my Chaotix 3D Flyer

2S Battery Example: \$3.95 on special and shipped from NSW, Australia

Turnigy nano-tech 300mAh 2S 35~70C Lipo Pack

<https://hobbyking.com/en-us/turnigy-nano-tech-300mah-2s-35-70c-lipo-pack.html>